

PREDICTING SMARTPHONE ADDICTION

USING COMPUTATIONAL INTELLIGENCE

Module	CIS6005 Computational Intelligence
Assessment	WRIT1 - Deep Learning Plus AI Mini Project
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Educational machine-learning prototype - not a medical or psychological diagnostic tool

Table of Contents

Table of Contents	2
List of Figures	3
List of Tables	3
1. Introduction	4
2. Comprehensive Overview of Computational Intelligence (10 marks)	4
3. Critical Literature Review (20 marks)	4
3.1 Construct, measurement and behavioural evidence	4
3.2 Wellbeing, context and predictive modelling	5
4. Dataset and Exploratory Data Analysis (10 marks)	5
5. System Architecture and ML Methods (10 marks)	7
6. Model Evaluation, Implementation and Practical Demonstration (40 marks)	8
6.1 Environment and preprocessing	8
6.2 Validation and model comparison	8
6.3 Kaggle result and serialization	9
6.4 API, frontend and tests	10
6.5 Practical demonstration	10
7. Critical Evaluation and Deep-Learning Suitability (10 marks)	11
8. Conclusion	11
References	12
Appendix A. Evidence gaps requiring student action	13
Appendix B. Verified API request fields	13
Appendix C. AI-tool disclosure placeholder	13

List of Figures

Figure 1. Target class distribution

Figure 2. Numerical correlation matrix

Figure 3. Screen-time relationship with target

Figure 4. Inference architecture

Figure 5. LightGBM validation confusion matrix

Figure 6. LightGBM feature importance

Figure 7. Kaggle submission and leaderboard evidence

Figure 8. Working application prediction interface

List of Tables

Table 1. Literature comparison

Table 2. Dataset feature categories

Table 3. EDA findings and model decisions

Table 4. Model comparison

Table 5. Cross-validation results

Table 6. API endpoints

Table 7. Limitations and improvements

1. Introduction

Smartphone-use data combine behavioural intensity, context and self-reported wellbeing. This project develops a reproducible computational-intelligence (CI) workflow for the Kaggle competition Predicting Smartphone Addiction. Its aim is to compare classical, ensemble and neural models, select the strongest validated pipeline, and expose that model through a responsible web application. The objectives were to audit and preprocess the competition data; engineer defensible behavioural ratios; compare Logistic Regression, Random Forest, LightGBM and a multilayer perceptron (MLP); use stratified holdout and cross-validation; generate a Kaggle submission; serialize the complete pipeline; and implement a validated FastAPI and browser interface.

The scope is prediction of the competition label `addicted_label`, not clinical assessment. The system is an educational machine-learning prototype and not a medical or psychological diagnostic tool. It minimises requested data, rejects the Kaggle identifier, stores no application records, presents probability as uncertainty, and labels explanations as model influences rather than causes.

2. Comprehensive Overview of Computational Intelligence (10 marks)

Artificial intelligence is the broad ambition of producing computational behaviour associated with perception, reasoning or action. Traditional symbolic AI encodes explicit rules and representations: it can be auditable when domain rules are stable, but brittle where behaviour is noisy, incomplete and context dependent. Machine learning instead estimates functions from examples. Deep learning is a machine-learning family using layered representation learning, while CI emphasises adaptive, data-driven techniques that tolerate uncertainty and approximate rather than require a complete symbolic world model.

CI is therefore appropriate to smartphone-use prediction because fixed rules such as 'more than x hours equals addiction' cannot capture interactions among screen time, social media, sleep, notifications and context. Logistic Regression offers a transparent linear baseline; Random Forest reduces variance through bagged trees (Breiman, 2001); LightGBM sequentially corrects errors with efficient gradient-boosted trees (Ke et al., 2017); and the MLP tests learned nonlinear representations. Advantages are adaptability, scalable pattern discovery and probabilistic ranking. Disadvantages include dependence on representative labels, limited causal meaning, distribution shift and less transparent decisions. Unlike a rule engine, the fitted model generalises statistically but can reproduce dataset bias. Responsible CI therefore requires validation, uncertainty communication and human accountability, not performance alone.

3. Critical Literature Review (20 marks)

3.1 Construct, measurement and behavioural evidence

Terminology is contested. Harris et al. (2020) reviewed 78 problematic-use scales and found uneven theoretical foundations, internal consistency and test-retest evidence. This weakens direct comparison between studies and supports cautious use of the competition label. Ratan et al. (2021) found consistent health associations across 27 adult studies, but every included study was cross-sectional; association cannot establish that smartphone use caused the outcome. Wacks and Weinstein (2021) similarly synthesised links with sleep, mood and cognition, while emphasising co-occurrence rather than a validated diagnostic pathway.

Objective sensing can reduce recall error. Ryding and Kuss (2020) found passive measures centred on screen time and checking behaviour across 18 studies and argued that short monitoring periods can capture ecologically valid patterns. Lee and Kim (2021) used 29,712 respondents and 27 personal/log variables: Random Forest achieved 82.59% accuracy, exceeding their decision tree, while basic demographics contributed relatively little. By contrast, this project uses competition data with missing behavioural and demographic fields and optimises ROC AUC rather than accuracy. The comparison supports tree ensembles and detailed usage measures but does not make results directly equivalent.

3.2 Wellbeing, context and predictive modelling

Duan et al. (2021) analysed 3,615 Chinese children and adolescents during COVID-19; a decision tree reached AUC 0.884, with internet use, smartphone hours, anxiety, injury fear and sex selected as factors. Nikolic et al. (2023) studied 761 Serbian medical students and reported associations with more than four hours' daily use, sleep quality, stress, anxiety and depression. Sela, Rozenboim and Ben-Gal (2022), using 215 participants, distinguished aware from concurrent or late-night use and found different quality-of-life associations. These studies disagree on the importance of demographics and operationalise outcomes differently, indicating that context matters and that a single threshold is unsafe.

The repository includes screen time, social-media time, gaming, sleep, stress and work/academic impact, so the literature informed behavioural coverage and the sleep-deficit/ratio features. It also motivated privacy minimisation and non-diagnostic wording. It did not justify causal claims, clinical thresholds or treatment advice. From a methods perspective, benchmark evidence indicates that tree ensembles remain strong on medium-sized tabular problems with irregular decision boundaries (Grinsztajn, Oyallon and Varoquaux, 2022). Gorishniy et al. (2021) show that carefully designed tabular neural baselines can be competitive, so an MLP comparison is appropriate, but superiority should be demonstrated rather than assumed.

Table 1. Literature comparison

Study	Data/method	Main evidence	Limitation / design implication
Harris et al. (2020)	Systematic review; 78 scales	Measurement quality varies	Label is not a clinical ground truth
Ryding and Kuss (2020)	Systematic review; 18 passive-monitoring studies	Screen/check patterns can be measured objectively	Instrumentation and privacy challenges
Lee and Kim (2021)	29,712; DT, RF, XGBoost	RF accuracy 82.59%	Different data/metric; supports ensemble comparison
Duan et al. (2021)	3,615; regression and decision tree	Decision-tree AUC 0.884	Cross-sectional, pandemic-specific sample
Ratan et al. (2021)	Systematic review; 27 studies	Consistent physical/mental-health associations	All included studies cross-sectional
Nikolic et al. (2023)	761 medical students; regression	Use, sleep and distress associated	Restricted population; no causality
Grinsztajn et al. (2022)	Tabular benchmark	Trees remain strong on typical tabular data	Benchmark conclusion is task-dependent

4. Dataset and Exploratory Data Analysis (10 marks)

The competition training set contains 691,369 rows and 14 columns; test contains 296,302 rows and 13 columns. Training comprises ID, 12 raw predictors and the binary target. No duplicate rows or IDs were found. Class 1 contains 490,474 records (70.9424%) versus 200,895 class 0 records (29.0576%), making unstratified accuracy potentially misleading.

Table 2. Dataset feature categories

Category	Verified fields
Numeric	age; daily screen time; social media; gaming; work/study; sleep; notifications; app opens; weekend screen time
Categorical	gender; stress level; academic/work impact
Engineered	social-media ratio; gaming ratio; notifications per screen hour; absolute sleep deficit from 8 hours
Excluded	id (row identifier)
Target	addicted_label

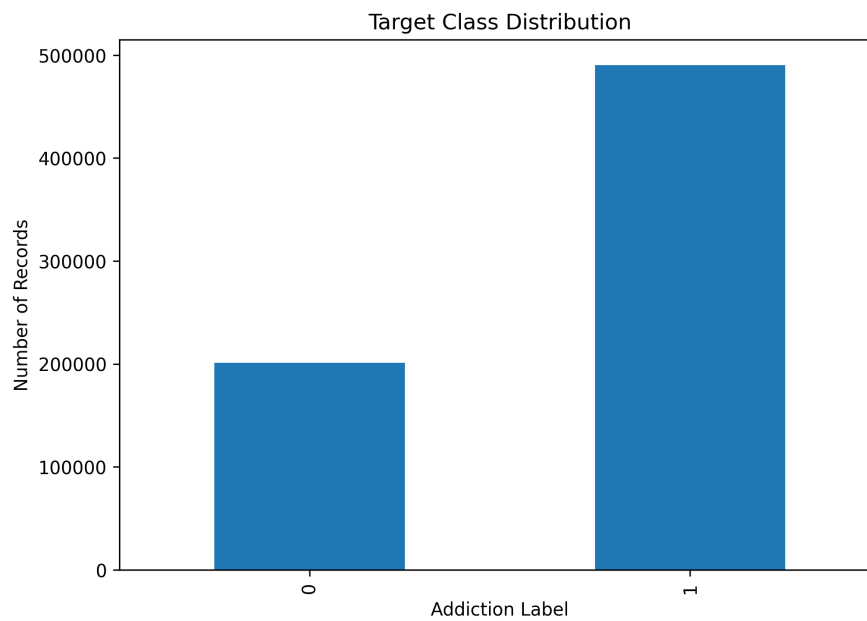


Figure 1. Target class distribution
Source: Author's analysis.

Figure 1 confirms imbalance. Missingness affects every raw predictor: 4.18% for age up to 19.38% for social-media hours; 422,184 rows contain at least one missing value. Deleting incomplete rows would discard substantial information and may distort the sample. Numeric train/test standardized mean differences are very small (absolute range 0.000035-0.002226), although matching means do not rule out local shift.

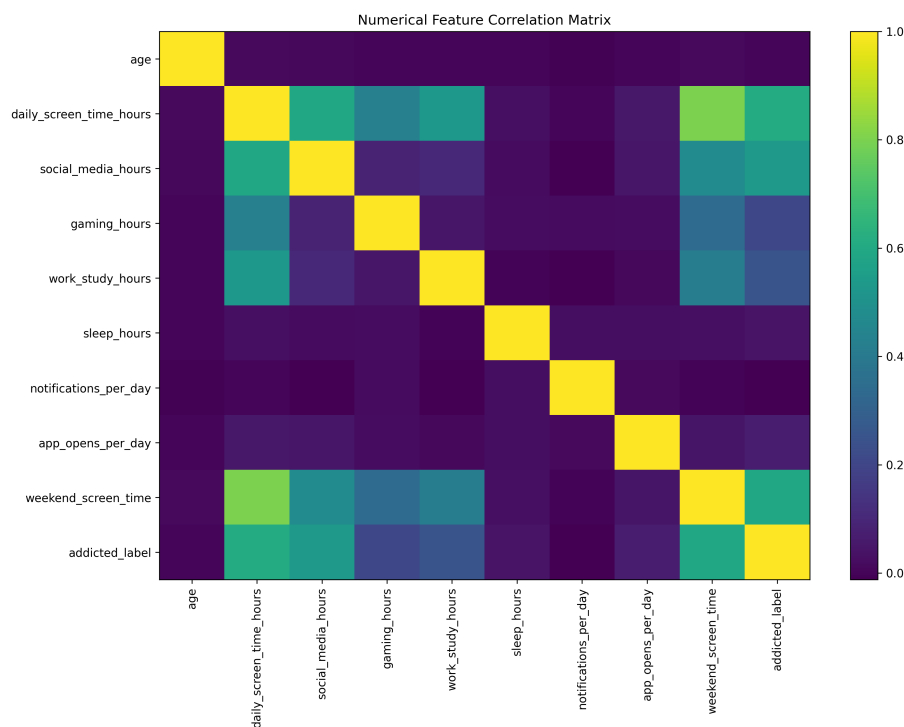


Figure 2. Numerical correlation matrix
Source: Author's analysis.

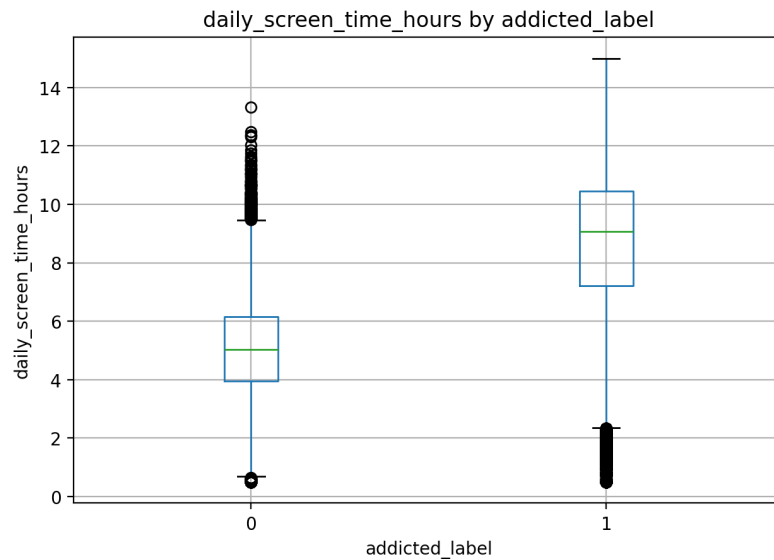


Figure 3. Daily screen time by target class
Source: Author's analysis.

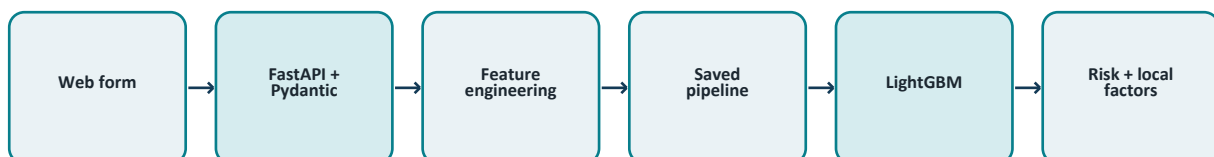
Figures 2 and 3 show correlated behavioural intensity and marked nonlinear class separation, particularly for daily screen time; social-media, gaming, work/study, app-opening and weekend-use plots show additional differences. This supports comparing boosting and bagging with linear and neural baselines, while avoiding the claim that any feature causes the label.

Table 3. EDA findings and model decisions

EDA finding	Observed evidence	Implemented decision
Class imbalance	70.94% class 1	Stratified split/CV; ROC AUC and balanced metrics
Substantial missingness	4.18%-19.38% by predictor	Median numeric and mode categorical imputation inside pipeline
Different scales	Hours, counts and age	Scale Logistic Regression/MLP; leave trees unscaled
Categorical inputs	Three object fields	One-hot encoding; unknown categories ignored
Nonlinear separation	Target boxplots and correlations	Compare RF/LightGBM with linear and MLP baselines

5. System Architecture and ML Methods (10 marks)

Training proceeds from Kaggle CSVs through EDA, deterministic feature engineering, pipeline preprocessing, a stratified holdout, four-model comparison, three-fold cross-validation of the top two models, full-data LightGBM fitting, joblib serialization and probability submission. Leakage is constrained because imputers, encoders and scalers are fitted inside each pipeline fold rather than before splitting.



Validated inference path - the fitted preprocessing is reused, not recreated

Figure 4. Online inference architecture.
Source: Author's design.

At inference, the browser obtains the model schema, posts 12 fields, and FastAPI applies Pydantic bounds and category checks. A one-row DataFrame is copied into shared feature engineering, producing 16 ordered

columns. The saved `ColumnTransformer` performs fitted preprocessing and LightGBM returns class probabilities. Application thresholds map class-1 probability to Low (<0.35), Moderate ($0.35\text{--}<0.65$) or High (≥ 0.65). Native LightGBM contribution values are grouped to readable factors; they are directional model influences, not causal percentages.

Logistic Regression provides a regularised linear, scaled baseline but cannot naturally express complex interactions. Random Forest averages decorrelated trees and is robust but trained more slowly here. LightGBM grows boosted trees sequentially, efficiently concentrating capacity on residual errors and nonlinear interactions. The MLP uses two hidden layers to learn nonlinear mappings, but requires scaling and tuning and is less transparent. The application differentiates itself through probability output, a validated API, shared preprocessing, local factor disclosure, privacy wording and reproducible artifacts; these are implementation characteristics, not a claim of research novelty.

6. Model Evaluation, Implementation and Practical Demonstration (40 marks)

6.1 Environment and preprocessing

The notebook was executed in Google Colab with Drive paths. Recorded training versions are Python 3.12.13, pandas 2.2.2, NumPy 2.0.2, scikit-learn 1.6.1 and joblib 1.5.3; the original LightGBM version was not recorded. The application pins FastAPI 0.115.9, Uvicorn 0.34.2, pydantic-settings 2.9.1 and LightGBM 4.7.0. Frontend technologies are HTML, CSS and JavaScript; pytest provides automated verification.

After separating `addicted\_label` and removing `id`, feature engineering creates social-media/screen-time, gaming/screen-time, notifications/screen-hour and absolute sleep-deficit features. Zero ratio denominators become missing. Numeric median and categorical most-frequent imputers, one-hot encoding and optional scaling live inside the pipeline. Scaling is used for Logistic Regression and MLP, not tree models. This design preserves identical fitted preprocessing when the joblib artifact is loaded for API inference.

6.2 Validation and model comparison

The 80/20 stratified split produced 553,095 training and 138,274 validation rows. Three-fold shuffled stratified cross-validation with random state 42 was then applied to LightGBM and Random Forest, the two strongest holdout models. ROC AUC is appropriate because Kaggle requires probability ranking and the target is imbalanced; accuracy at one threshold can hide minority-class errors. Precision, recall, balanced accuracy, F1 and log loss provide complementary views.

Table 4. Holdout model comparison

Model	Acc.	Bal. acc.	Precision	Recall	F1	ROC AUC	Log loss	Seconds
LightGBM	.8961	.8691	.9211	.9335	.9272	.9598	.2325	47.06
Random Forest	.8473	.8578	.9456	.8327	.8856	.9391	.3004	350.18
MLP	.8636	.8192	.8874	.9250	.9058	.9374	.2832	104.61
Logistic Reg.	.8334	.8355	.9271	.8305	.8761	.9147	.3819	303.96

Table 5. Three-fold ROC AUC

Model	Fold 1	Fold 2	Fold 3	Mean	SD
LightGBM	0.959698	0.960774	0.960673	0.960381	0.000485
Random Forest	0.939074	0.940262	0.940061	0.939799	0.000519

LightGBM achieved the highest holdout ROC AUC, accuracy, balanced accuracy and F1, the lowest log loss, and a stable three-fold mean. It also trained substantially faster than Random Forest in this run. These results, plus compact tabular inference, justify its selection. The final model used 600 estimators, learning rate 0.04, 31 leaves, minimum child samples 30, subsample and column sample 0.85, L1 0.1 and L2 0.2, then fitted all 691,369 rows in 62.75 seconds.

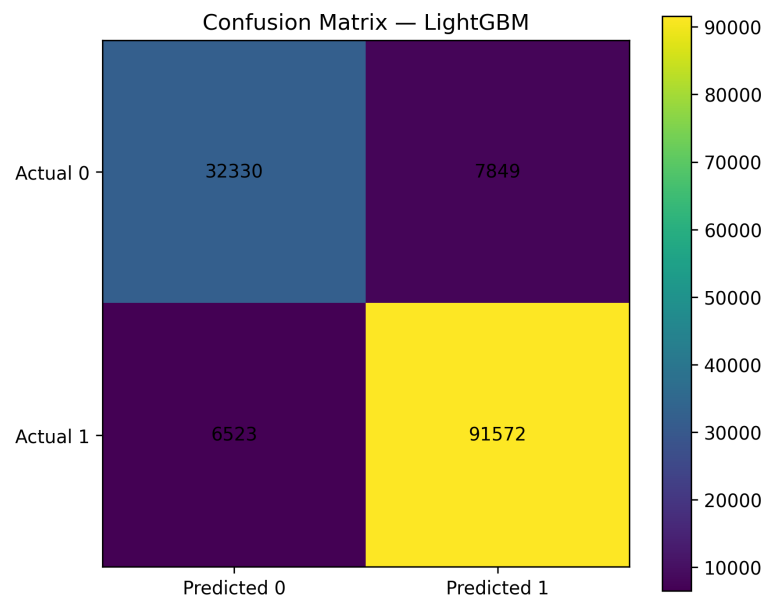


Figure 5. LightGBM holdout confusion matrix
Source: Author's analysis.

Figure 5 records 32,330 true negatives, 7,849 false positives, 6,523 false negatives and 91,572 true positives. False predictions matter because a high-risk display can worry a user, while a false negative can create false reassurance. The interface therefore avoids diagnosis and presents both probability and limitations.

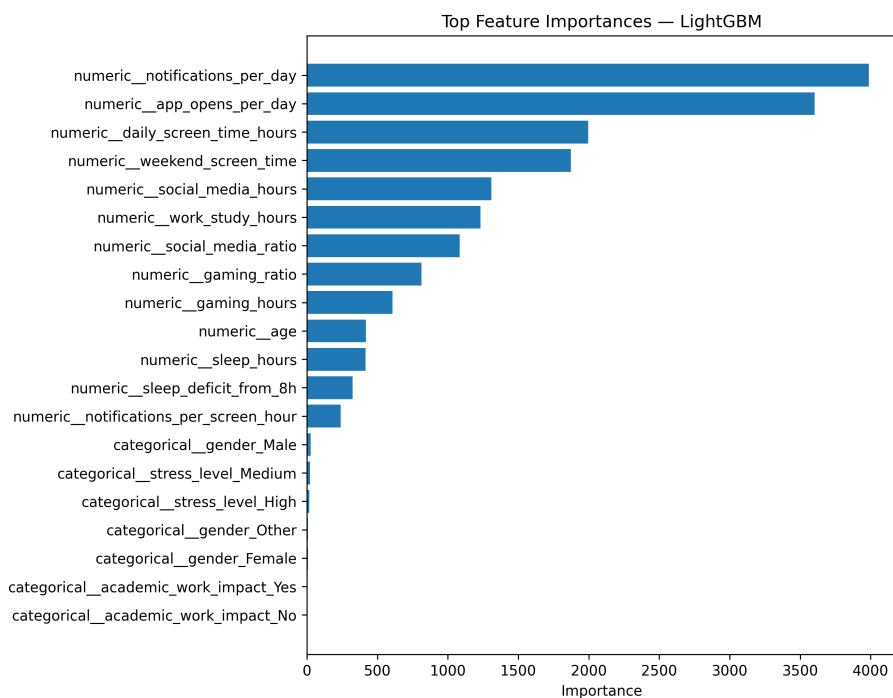


Figure 6. LightGBM split importance
Source: Author's analysis.

Notifications and app opens lead split importance, followed by daily and weekend screen time. Split counts are global model-use indicators, not causal effects. The API's local contribution section is separately labelled and limits output to five factors (Lundberg and Lee, 2017).

6.3 Kaggle result and serialization

The generated `submission\_lightgbm.csv` contains 296,302 test IDs and finite probabilities from 0.000713 to 0.999993. The provided verified public score is 0.96189 ROC AUC. This is competition ranking performance, not clinical accuracy. Public rank, competition-date eligibility, selected final/private submission and private score are not evidenced in the repository and are not claimed.

[STUDENT TO INSERT VERIFIED EVIDENCE] Public submission history, score 0.96189, competition dates, and selected final/private submission.

Figure 7. Kaggle submission and leaderboard evidence
Source: Student evidence required.

The final pipeline is serialized with joblib alongside metadata, environment versions and model requirements. API startup verifies required files, parses metadata, checks package compatibility, loads only the trusted local artifact, confirms ``predict``, ``predict_proba``, classes ``[0,1]`` and exact feature order, then stores one singleton in application state. Exact scikit-learn/joblib compatibility is important because pickle-based artifacts depend on Python class structure.

6.4 API, frontend and tests

Table 6. Implemented API endpoints

Method	Endpoint	Function
GET	/api/health	Status, environment, version and timestamp
GET	/api/model/info	Allow-listed model metadata
GET	/api/model/schema	Twelve raw form fields and categories
POST	/api/predict	Validation, probability, band, disclaimer and local factors

``api/main.py`` configures lifespan, CORS, middleware, routes and static serving. ``PredictionRequest`` in ``api/schemas.py`` forbids extra fields, NaN, infinity, invalid categories and out-of-range values. ``predict_addition`` builds the DataFrame, calls ``add_domain_features``, checks order, calls ``predict_proba``, resolves positive class 1 from schema, validates probability bounds, assigns the display band and requests a local explanation. Exceptions return safe JSON without paths or traces. Requests receive a UUID, size limit, structured log and defensive headers.

The frontend fetches schema dynamically, renders numeric/select controls with labels and helper/error associations, prevents duplicate submission, calls ``/api/predict``, and reveals an accessible SVG gauge only after success. It highlights one band, prints the model message/version/disclaimer, and displays local influences with a fallback. Responsive CSS, visible focus and reduced-motion support improve access. No login or application database is implemented.

The final audit executed 39 pytest tests successfully. Coverage includes feature order and immutability; zero denominators; artifact and version failures; singleton load; health/static routes; missing, extra, categorical, numeric and non-finite validation; low/moderate/high mapping; real-model integration; probability bounds; safe 500 responses; local explanation structure/fallback; request size, IDs, headers and CORS. ``compileall`` also passed. Docker configuration is present, but its build is not claimed because Docker was unavailable in the verification environment.

6.5 Practical demonstration

The demonstrable flow is: start Uvicorn; open the landing page; navigate to the prediction form; submit the 12 fields; observe Pydantic validation, feature engineering and singleton pipeline inference; review probability, band, explanation and disclaimer; then submit an invalid value to show HTTP 422 feedback. Swagger, ``/api/health`` and ``/api/model/info`` expose the operational contract without local paths. Programmatic HTTP checks passed for these routes and a real prediction, but screenshot evidence remains to be inserted.

[STUDENT TO INSERT VERIFIED EVIDENCE] Landing page, populated form, successful result, validation error, Swagger, health, model info and test output.

Figure 8. Working application prediction interface
Source: Student evidence required.

7. Critical Evaluation and Deep-Learning Suitability (10 marks)

Strengths are the large competition sample, stable LightGBM cross-validation, full leakage-aware pipeline, exact inference feature order, safe model lifecycle, usable web/API interface and transparent limitations. The public ROC AUC is strong competition evidence. Nevertheless, the data may be synthetic and is not externally or clinically validated. Self-report and label measurement may embed error; Harris et al. (2020) show that problematic-use scales themselves vary. Demographic fairness was not evaluated, thresholds are arbitrary display choices, and probability calibration was not measured. Distribution shift, false positives, false negatives, privacy and automation bias remain material risks.

Deep learning is not automatically preferable. The MLP underperformed LightGBM here (ROC AUC 0.9374 versus 0.9598), required scaling and trained more slowly. The problem has only 16 engineered tabular columns, where tree-based models efficiently learn thresholds and interactions; benchmark research similarly finds tree ensembles difficult to displace on typical tabular tasks (Grinsztajn, Oyallon and Varoquaux, 2022). Gorishniy et al. (2021) caution that strong neural designs and tuning can narrow the gap, so this result rejects this MLP configuration, not deep learning generally.

Deep architectures become more plausible with longitudinal event sequences, app-level logs, text or sensor streams. Recurrent networks or temporal transformers could model changing behaviour, but require consent, larger storage, stronger privacy controls and temporal/external validation. Priorities are validation on real diverse cohorts; subgroup fairness/error analysis; probability calibration and threshold optimisation; model/drift monitoring; stronger causal-study design; and privacy-preserving or federated learning. SHAP-style local explanations should be user-tested rather than assumed beneficial. NIST's AI Risk Management Framework emphasises governance, measurement and ongoing management (NIST, 2023), aligning with consent, minimisation, transparency and accountable deployment rather than an unsupported claim of legal compliance.

Table 7. Limitations and evidence-based improvements

Limitation	Risk	Improvement
Competition/possibly synthetic data	Weak real-world generalisation	External longitudinal validation
No fairness audit	Unequal subgroup errors	Report subgroup ROC AUC, calibration and error rates
Uncalibrated display thresholds	Misleading risk communication	Calibration and decision-specific threshold study
Cross-sectional behavioural inputs	No temporal or causal inference	Consent-based sequence modelling and causal designs
No deployment monitoring	Silent drift	Input/performance monitoring and review triggers
Sensitive behavioural data	Privacy and stigma	Minimisation, retention controls, federated learning research

8. Conclusion

This project compared linear, bagged-tree, boosted-tree and neural approaches for competition smartphone-addiction prediction. Leakage-aware preprocessing, four verified engineered features, stratified validation and multiple metrics supported selection of LightGBM. It achieved holdout ROC AUC 0.9598, three-fold mean 0.9604 and the provided public Kaggle score 0.96189, then powered a validated FastAPI and accessible web interface. The principal contribution is a reproducible route from notebook evidence to responsible probability delivery; the principal limitation is the absence of external clinical validation and complete Kaggle final/private evidence. Future work should prioritise real longitudinal data, fairness, calibration, privacy and drift monitoring before considering more complex temporal deep learning. The output remains educational, uncertain and non-diagnostic.

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Appendix A. Evidence gaps requiring student action

- [STUDENT TO INSERT VERIFIED EVIDENCE] Competition Overview with verified start/end dates.
- [STUDENT TO INSERT VERIFIED EVIDENCE] Kaggle Evaluation page confirming ROC AUC.
- [STUDENT TO INSERT VERIFIED EVIDENCE] Public submission history and leaderboard screenshot linking score 0.96189.
- [STUDENT TO INSERT VERIFIED EVIDENCE] Selected final/private submission and private result when available.
- [STUDENT TO INSERT VERIFIED EVIDENCE] Application, API, Swagger and pytest screenshots.
- [STUDENT TO INSERT VERIFIED EVIDENCE] Moodle deadline and applicable AI-use disclosure requirement.

Appendix B. Verified API request fields

Required raw fields: age; daily_screen_time_hours; social_media_hours; gaming_hours; work_study_hours; sleep_hours; notifications_per_day; app_opens_per_day; weekend_screen_time; gender; stress_level; academic_work_impact. The ID field is excluded.

Appendix C. AI-tool disclosure placeholder

[STUDENT TO INSERT VERIFIED EVIDENCE] Add the university-approved disclosure of AI assistance after checking the current Moodle and institutional policy. The student must review every claim and citation and remain responsible for the submitted work.